

Final Exam

Applied Optimization

Procedures:

Leave your **student ID card** on the table during the examination for verification. Write your **first and last name** as well as your **student number** on all answer sheets. Use only black or blue ink pens and write legibly. Exam questions done in pencil or other colored pens will **NOT** be graded. Check if all **9 pages** (5 sheets) are present in your exam copy.

Regulations:

Only one sheet of A4 paper with handwritten notes on both sides is permitted. Course notes, textbooks, other books/printed materials, calculators are **NOT** permitted during the examination. The use of electronic devices such as cell phones, PDAs, laptops, MP3 players, iPods, etc. is strictly **PROHIBITED** during the examination.

The student is deemed to have failed the course if they are found cheating or violating any of the above regulations.

NOTE: You do not necessarily need to finish the whole exam to obtain the maximal grade.

<i>Exercise</i>	<i>Points</i>	<i>Earned points</i>
1. Convex Sets	8	
2. Convex Problems	14	
3. Line Search - Part I	16	
4. Line Search - Part II	6	
5. Trust-Region and Active Set Methods	26	
6. Algorithms	16	
7. Magic Ball System	14	
8. Quasi-Newton's method	10	
Total	110	

1 Convex Sets (8 points)

1.1. (4 points): Sketch the following domains and indicate if they are convex or not

$$S_a = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

$$S_b = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1, xy \leq 0\}$$

1.2. (2 points) Assume two *convex* sets A and B . Is $A \cap B$ also convex? If yes, justify your answer with a proof, otherwise, sketch a counter-example.

1.3. (2 points) What about $A \cup B$? If yes, justify your answer with a proof, otherwise, sketch a counter-example.

2 Convex Problems (14 points)

You want to solve the following problem:

$$\begin{aligned} \min \quad & f(x, y) = -\ln(y) \\ \text{subject to} \quad & x^2 + y^2 \geq 1 \\ & x^2 + y^2 \leq 4 \\ & x, y \geq 1 \end{aligned}$$

where $f : \mathbb{R}_{>0}^2 \rightarrow \mathbb{R}$

- 2.1. (2 points) Is the objective function convex? Justify your answer.
- 2.2. (2 points) Is this problem a convex problem? Justify your answer.
- 2.3. (2 points) Sketch the feasible domain and mark the optimal solution with a circle.
- 2.4. (6 points) Verify the optimality of your solution with the KKT conditions.
Note: If you did not answer to the previous question, you can use any feasible point and analyze the KKT conditions at this point.
- 2.5. (2 points) Transform the problem into any constrained, *convex* problem with the same optimal solution. The feasible domain should be more than a single point.

3 Line Search - Part I (16 points)

- 3.1. (2 points) Given the objective function $f_a(x, y) = \frac{1}{12}(x^4 + y^4) - a(x^2 + y^2)$, is there any value of a for which $f_a(x, y)$ is convex on $\mathbb{R}^2$? Justify your answer.

Let us now set $a = 8$, giving $f_8(x, y) = \frac{1}{12}(x^4 + y^4) - 8(x^2 + y^2)$.

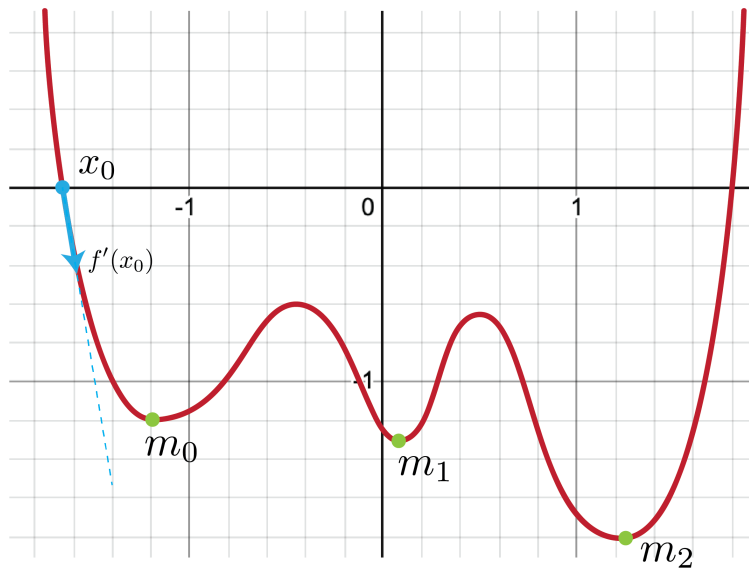
- 3.2. (4 points) What is the minimum of f_8 , restricted to a line in the direction $v_0 = (0, 1)^T$ and starting from $p_0 = (0, 0)^T$? We will call this minimum p_1 .

If you did not find p_1 you can use $p_1 = (0, 7)^T$ for the rest of the exercise.

- 3.3. (2 points) What would be the issue in using p_1 as a starting point of the standard version of Newton's method?
- 3.4. (8 points) Starting from p_1 and in search direction $v_1 = (1, 0)^T$, find a point p_2 that could be used as a starting point that would guarantee the standard version of Newton's method to work.

4 Line Search - Part II (6 points)

Let us now consider the 1-dimensional smooth function plotted below, with its three local minima m_0, m_1 and m_2 .



- 4.1. (3 points) Starting from point x_0 on the plot, is there any value of α for which all three minima m_i satisfy the Armijo condition? If yes, draw a line representing the Armijo condition on the plot. Otherwise, justify why not.
- 4.2. (3 points) Still starting from point p_0 , could we use the strong Wolfe conditions to limit the candidates to m_0 and m_2 ? Justify your answer (you do not need to provide any value).

5 Active Set Trust Region Method (26 points)

In this task, we want to solve the following problem:

$$\text{minimize } f(x,y) = \frac{1}{3}x^3 - x \cos^2(y)$$

To do so, we will modify the Trust Region method to work with the 1-norm instead of the 2-norm. This will make each subproblem a quadratic problem with linear constraints, instead of quadratic constraints. We will thus apply the Active Set method to solve each subproblem.

- 5.1. (4 points): Rewrite the 1-norm constraint formulation as a set of 4 linear inequality constraints. Each of them represents a line spanned by an edge of the 1-norm ball (rombus) of radius $\delta_0 = \frac{1}{2}$. As a reminder, the 1-norm is defined as $\|v\|_1 = \sum_{i=1}^n |v_i|$, where v_i is the i -th component of $v \in \mathbb{R}^n$.
- 5.2. (3 points): Compute the gradient and the Hessian of the function.
- 5.3. (4 points): Express the TR subproblem at the starting point $p_0 = (2, \pi)^T$.
- 5.4. (8 points): Assuming the empty set to be the initial set of active constraints, solve the TR subproblem with Active Set Method. To obtain a search direction, use Newton's method with equality constraints. Write the intermediate steps of the computations in the table below. *Hint*: no modification of the Hessian is required.

k	p_k	A_k	p_{k+1}	λ_{k+1}	A_{k+1}	KKT conditions satisfied (yes/no)

- 5.5. (3 points): Sketch the optimization path inside the boundary rombus for the current subproblem.
- 5.6. (4 points): Compute the relative reduction and argue whether the Trust Region radius δ should be enlarged or shrunk. In case you failed to compute the minimum of the subproblem, simply write down the condition and explain it with your own words.

6 Algorithms (16 points)

Consider the task of optimizing a problem $f(x)$, $x \in \mathbb{R}^n$, $n = 10^6$ with $f \in C^d$ (i.e. d times continuously differentiable over $\mathbb{R}^n$), with m independent linear equality constraints and with or without a feasible starting point x_0 . The following algorithms are available, as implemented in the lecture:

- (a) L-BFGS
- (b) Feasible start Newton's method with linear equality constraints
- (c) Quadratic penalty method with Gradient Descent
- (d) Active-Set method
- (e) Big M method
- (f) Sequential Quadratic Programming

Note: algorithms (d), (e) and (f) all use feasible start Newton's method with linear equality constraints and Hessian projection as their underlying optimization method.

6.1. (12 points) For each set of parameters given in the table below, give the list of suitable methods to obtain a (local) minimum.

Note: You will get -1 point every time you give one algorithm that is actually not suitable (your total for **6.1** cannot go below 0 points).

	f convex	sparse	m	C^d	feasible x_0	suitable methods
1	yes	yes	10^5	C^∞	yes	
2	no	no	10^5	C^∞	no	
3	no	yes	0	C^1	yes	
4	yes	yes	10	C^1	yes	

6.2. (4 points) For each problem, indicate which method you would use to be most efficient and *briefly* justify why.

BONUS (2 points) Which algorithm would you use to find the global optimum for a non-convex, dense problem with $n = 10^9$, $m = 10^9$, $f \in C^\infty$ and a feasible starting point x_0 ? Justify your answer.

7 Magic Ball System (14 points)

In this task, we will handle a similar problem to the Mass Spring System we saw in the exercises. Instead of masses connected by springs, we will consider a set of 10^6 magic balls that all interact with each other. Specifically, they want to be at a distance of r to *all* others. We will assume that this is a 2D problem for simplicity (magic balls are constrained to a plane) and we want to design an optimization problem to find the system's equilibrium.

- 7.1. (3 points) For a given pair of magic balls at positions p_a and p_b , how would you design a *differentiable* energy that is zero when the distance between the two magic balls is r , and positive when they are closer or further apart?
- 7.2. (5 points) First, we want to solve the problem with Gradient Descent. You have to assemble the global gradient of the objective function, by filling the missing part of the code below (feel free to use the next page if you need more space).

```
using Vec2d = Eigen::Vector2d;
using Vec4d = Eigen::Vector4d;
using Vec = Eigen::VectorXd;

//computes the gradient of the energy between two masses a and b
void grad_f(Vec2d pos_a, Vec2d pos_b, Vec4d& grad);

//computes the gradient of the whole problem.
//positions are stored as (px_0, py_0, px_1, py_1,...) and the gradient as (gx_0,
//gy_0, gx_1, gy_1, ...)
void grad_problem(const Vec& positions, Vec& grad){

    grad.setZero();
    for(int i(0); i<(int)positions.size(); i++){
        for(int j(i+1); j<(int)positions.size(); j++){

            /** YOUR CODE HERE **/

        }
    }
}
```

- 7.3. (2 points) Assuming we projected the Hessian to make it positive semidefinite, do you think it would improve performance to use Newton's Method in this context? Justify your answer.
- 7.4. (2 points) Now let's say that at least one magic ball wants to stay within the neighborhood of the origin. How would you modify the problem accordingly? You do not need to modify the code.
- 7.5. (2 points) Now, we want to modify the energy so that magic balls only interact with each other if they are *currently* within a distance of $3r$. Modify the code to compute the gradient of this new energy.

BONUS (4 points) With the problem modified as in 7.5, would you change your answer to 7.3? Justify your decision.

8 Quasi-Newton's method (10 points)

Assume that you want to optimize a nonlinear problem with a Quasi-Newton's method. As you know from the lecture, the key of the QN methods is to learn the Hessian on the fly, based on the history of gradient information and the function values.

Assume that you know the following:

- A sequence of optimization points: $x_0 = (-1, 1)^T, x_1 = (0, 0)^T, x_2 = (0, 1)^T$
- Objective function values at these points: $f(x_0) = 4, f(x_1) = 3, f(x_2) = 2,$
- Gradients at these points: $\nabla f(x_0) = (0, 1)^T, \nabla f(x_1) = (1, 0)^T, \nabla f(x_2) = (2, 2)^T$

8.1. (4 points): Compute a Hessian approximation $B_1 \in \mathbb{R}^{2 \times 2}$ such that the gradients of the last two iterates are interpolated.

8.2. (6 points): Compute the Hessian approximation $B_2 \in \mathbb{R}^{2 \times 2}$ such that the gradients of the last three iterates are interpolated.

BONUS (2 points): Is it always possible in $\mathbb{R}^2$ to find a Hessian approximation, where interpolation of three subsequent gradients is ensured, like in **8.2**? Justify your answer, and identify additional conditions if necessary.